

Joaquin Vanschoren, Ph.D.

✉ joaquin.vanschoren@gmail.com

☎ +32 497 903069

🏠 Genk, Belgium

🌐 <https://joaquinvanschoren.github.io>

🐦 @joavanschoren

🌀 joaquinvanschoren

Profile

I aim to scientifically understand and build AI systems with advanced capabilities and make AI open and accessible to benefit all of humanity. I founded OpenML to build a global open machine learning infrastructure, started the NeurIPS Evaluations and Datasets track to incentivize better data and evaluations, and created new machine learning standards (e.g. Croissant) to connect communities. I gave tutorials at top AI conferences (e.g., NeurIPS), worked with top AI research labs (e.g., Google Deepmind), advice on AI policy (e.g., EU AI Office), and have been interviewed for top news articles (e.g. Nature, Science) and podcasts.

Professional Experience

- 1/2023 - present 📌 **Associate Professor, TU Eindhoven**, Netherlands.
- Leading a thriving AI research lab (20+ PhDs, postdocs, AI Engineers).
 - Presented tutorials at major conferences (e.g. NeurIPS, AAAI), and 30+ invited talks.
 - Attracted 25+ grants (EU, DARPA, NWO, ...) with €8+ million for my own group.
 - Initiated and chaired the Evaluations and Datasets Track at NeurIPS.
 - Chairman OpenML Foundation & Co-chair for MLCommons Working Groups.
 - Education Director for the Data Science and AI program.
 - Best Teacher award for *Machine Learning Engineering* course, 2.3k YouTube subs.
 - Expert for European Commission JRC panel on the European AI Act.
- 5/2025 - 8/2025 📌 **Visiting Faculty Researcher, Google DeepMind**, Paris, France.
- Developed Eclair, an MCP framework to help AI agents work with datasets.
- 1/2014 - 12/2022 📌 **Assistant Professor, TU Eindhoven**, Netherlands.
- Founded the *OpenML* open-source project, with currently 11 core developers, 300k users, and integrations in key ML frameworks and languages.
 - Co-authored book on AutoML: 1M downloads, 2.5k citations, 4.5/5 Amazon rating.
 - Winner of an Amazon Research Award and the Dutch Data Prize
 - Founding member of European AI societies ELLIS and CLAIRE
- 1/2013 - 12/2013 📌 **Data Scientist, CityLife** (now: Joyn), Belgium.
- Developed and optimized a recommender system for 150k users.
- 9/2010 - 8/2013 📌 **Post-doctoral Fellow and Lecturer, Leiden University**, Netherlands.
- 5/2010 - 9/2010 📌 **Post-doctoral Fellow, KU Leuven**, Belgium.
- 8/2005 - 5/2010 📌 **Ph.D. researcher, KU Leuven**, Belgium, and **Waikato University**, New Zealand.
- Won an ECML best demo award. Published original research in meta-learning.

Education

- 2024 – 2025 📌 **Educational Leadership Programme, Utrecht University (EWUU Alliance)**
- 2014 – 2016 📌 **Dutch University Teaching Qualification (UTQ)**
- 2005 – 2010 📌 **Ph.D. Computer Science, KU Leuven.**
- Thesis title: *Understanding Machine Learning Performance with Experiment Databases.*
Advisors: Hendrik Blockeel and Geoffrey Holmes (Univ. Waikato, New Zealand)
- 2000 – 2005 📌 **M.Sc. Computer Science, KU Leuven, cum laude.**
- Thesis title: *A framework for high-level perception*, magna cum laude.
Advised by Prof. Douglas R. Hofstadter, Indiana University in Bloomington.

Awards and Fellowships

2025	■	Patterns Editor Pick. Top-5 article of 2025 in the journal Patterns (Cell Press)
2024	■	NeurIPS Spotlight paper. Top 3% of submissions at NeurIPS 2024
	■	ICML Spotlight paper. Top 3% of submissions at ICML 2024
2022	■	Best Teacher award. Dutch study association for data science students (Pattern)
2019	■	ELLIS Founding member. European Laboratory for Learning and Intelligent Systems
	■	Amazon Research Award, Amazon Research
2016,2017	■	Microsoft Azure Research Award, Microsoft Research
2016	■	Dutch Data Prize, Research Data Netherlands
2009	■	Best Demo Award, 17th European Conference on Machine Learning (ECML-PKDD)

In the Media

1/5/2024	■	<i>Royal Society</i> - Science in the age of AI.
16/4/2024	■	<i>IEEE Spectrum</i> - Announcing a Benchmark to Improve AI Safety.
5/12/2023	■	<i>Nature</i> vol 624, issue 7990 - Is AI leading to a reproducibility crisis in science?
6/5/2022	■	<i>Science</i> vol 376, issue 6593 - Taught to the test. AI software clears high hurdles on IQ tests but still makes dumb mistakes. Can better benchmarks help?
5/5/2021	■	<i>The Biotech podcast</i> , Season 1, Episode 20 - Dr Joaquin Vanschoren on Making Data Public
5/2020	■	<i>KDnuggets</i> - Automated Machine Learning: The Free eBook
13/4/2020	■	<i>Science</i> - Artificial intelligence is evolving all by itself
16/2/2018	■	<i>Science</i> vol 365, issue 6451 -Artificial intelligence faces reproducibility crisis.
10/2016	■	<i>Open Science Radio podcast</i> , Episode 59 - OpenML
8/2014	■	<i>KDnuggets</i> - OpenML: Share, Discover and Do Machine Learning

Skills

Leadership	■	Head of the Advanced Models by Open Research & Engineering (AMORE) research lab, with 20+ researchers (PhDs, postdocs, AI Engineers). Director of the TU Eindhoven Data Science program. Open source project lead (OpenML). MLCommons working group chair (AI Safety and Croissant). Conference chair (Track Chair @ NeurIPS 2021-2026, Program Chair @ AutoML 2024 and DS 2018, General Chair @ LION 2016).
Academic	■	Published 200+ papers, including top journals and conferences. Several best paper awards and spotlight papers. Taught tutorials and summer schools at major venues (NeurIPS, AAAI, ACDL,...). Gave 30+ invited talks. Edited and reviewed for major journals and conferences (JMLR, NeurIPS, ICML,...).
Coding	■	Proficient in Python, Javascript. Experience with R, Java. Open-source development. AI-assisted development.
Technologies	■	ML libraries (PyTorch, TensorFlow, scikit-learn,...), MLOps (TensorBoard, MLFlow, W&B, OpenML), scalable GPU training (NVIDIA), databases (MySQL, Postgres), search engines (ElasticSearch), web frameworks (Next.js, React, ...), API development (FastAPI), server management (Linux, K8s).

Teaching Experience

University courses

- 2019-present  **Machine Learning Engineering (M.Sc, 250 students)**, TU Eindhoven. Evaluation: 8.8/10. Teaching award ('pluim'), Best Teacher award from the Dutch study association for data science students (DSA Pattern) in 2022, and nominated as Best Teacher of TU/e in 2022 and Data Science in 2026.
- 2018-2023  **Data Mining (M.Sc, 75 students)**, Jhieronimus Academy of Data Science. Evaluation: 8.1/10. Teaching award ('pluim').
- 2016-2017  **Data Mining (B.Sc, 140 students)**, Tilburg University. Evaluation: 8.4/10. Teaching award ('pluim').
- 2015-2017  **Foundations of Data Mining (M.Sc, 80 students)**, TU Eindhoven. Evaluation: 7.8/10
- 2014-2015  **Web-scale Information Systems (M.Sc, 60 students)**, TU Eindhoven. Evaluation: 7.7/10
- 2014-2017  **Web Technology (B.Sc, 80 students)**, TU Eindhoven. Evaluation: 7.2/10
- 2011-2014  **Data Mining (B.Sc, 60 students)**, Leiden University. Evaluation: 7.6/10

Invited Lectures

- 2026  **International Programme on AI Evaluation** Science of Benchmarking, Safety evaluations. Cambridge, UK and Valencia, Spain.
- 2025  **CCAIM AI and Machine Learning Summer School** Auto-Continual Learning. Cambridge, UK.
-  **ELLIS-CISPA Summer School on Trustworthy AI** AI Safety Benchmarks. Saarbrücken, Germany.
-  **Boston University** Auto-Continual Learning. Boston, USA.
- 2023  **AutoML Fall School 2023**. Metalearning for AutoML. Munich, Germany.
- 2022  **ACDL 2022**. AutoML (3 lectures). Advanced Course on Data science and Machine Learning, Pontigniano, Italy.
- 2021  **Univ. Trento** AutoML lecture. Advanced topics in ML and Optimisation, Trento, Italy.
-  **ACDL 2021** Metalearning (3 lectures). Advanced Course on Data science and Machine Learning, Pontigniano, Italy.
- 2019  **ACDL 2019** AutoML (3 lectures). Advanced Course on Data science and Machine Learning, Pontigniano, Italy.
- 2017  **Geilo Winter School 2017** Tutorial on Machine Learning, Geilo, Norway.

Tutorials

- 2021  **AAAI 2021** Tutorial on Metalearning. AAAI Conference on Artificial Intelligence.
-  **DSAA 2021** Tutorial on Metalearning. Data Science and Advanced Analytics Conference.
-  **ODSC Europe 2021** Tutorial on AutoML. Open Data Science Conference.
- 2019  **ODSC Europe 2019** Tutorial on AutoML. Open Data Science Conference.
- 2018  **NeurIPS 2018** Tutorial on Automated Machine Learning, with Frank Hutter. Neural Information Processing Systems.
- 2017  **ECMLPKDD 2017** Tutorial on Automated Machine Learning.
- 2015  **ECMLPKDD 2015** Tutorial on Metalearning and Algorithm Selection.

Leadership Experience

Academic Leadership training

- 2024-2024 Academic Leadership for Associate Professors, TU Eindhoven
- 2020-2020 Academic Leadership for Assistant Professors, TU Eindhoven

PhD Student Advisor

- 2026-... Luca Mainardi (NXP and TU Eindhoven). AI for semiconductor production.
- 2025-... Qingren Yao (TU Eindhoven). Multimodal foundation models.
- Shreya Sajid (TU Eindhoven). Multimodal foundation models.
- Fabian Denoodt (TU Eindhoven). Safety-Critical Multimodal Learning.
- Shawon Ashraf (TU Eindhoven). Open multilingual foundation models.
- Diana Alexandra Onutu (TU Eindhoven). Open multilingual foundation models.
- Dalton Harmsen (TU Eindhoven). Open multilingual foundation models.
- 2022-... Prabhant Singh (TU Eindhoven). Transfer Learning.
- 2021-... Pan Jiarong (TU Eindhoven). Bayesian Optimization and meta-learning.
- Fangqin Zhou (TU Eindhoven). Transformer architectures for vision.
- Andrei Simion-Constantinescu (TU Eindhoven). Self-supervised learning for vision.
- Elif Ceren Gok (TU Eindhoven). Automated continual learning.
- Murat Onur Yildirim (TU Eindhoven). Efficient and scalable continual learning.
- 2018-... Bilge Celik (TU Eindhoven). AutoML for evolving data.
- 2021-2026 Israel Campero Jurado (TU Eindhoven). AutoML and metalearning for time series.
- 2017-2022 Pieter Gijbbers (TU Eindhoven, MCS). Systems for AutoML research.
- 2015-2019 Chao Zhang (TU Eindhoven, IEIS, co-advisor). Data analysis for digital health.
- 2014-2018 Rafael Mantovani (Univ. Sao Paulo, ICMC Sao Carlos, co-advisor). Metalearning for hyperparameter tuning.
- 2012-2016 Jan van Rijn (Leiden University, LIACS). Massively collaborative machine learning.

Postdoctoral Fellow Advisor

- 2025-... Jie Liu, Post-Doc (TU Eindhoven).
- Thilina Rajapakse, Post-Doc (TU Eindhoven).
- Anna Vettoruzzo, Post-Doc (TU Eindhoven).
- 2023-... Alexis Cvetkov-Iliev, Post-Doc (TU Eindhoven).
- 2025-2025 Gennaro Gala, Post-Doc (TU Eindhoven).
- Matthew Danish, Post-Doc (TU Eindhoven).
- 2022-2024 Mert Kiliçkaya, Post-Doc (TU Eindhoven).
- 2019-2021 Marcos L.P. Bueno, Post-Doc (TU Eindhoven).

Research Engineer Advisor

- 2026-... Omosola Odetunde, AI Engineer, Machine learning (TU Eindhoven).
- 2025-... Ivan Slobozhan, AI Engineer, Multimodal Foundation Models (TU Eindhoven).
- Mohamed Salah Mahmoud, AI Engineer, Large Language Models (TU Eindhoven).
- 2024-... Subhaditya Mukherjee, AI Engineer, Machine learning (TU Eindhoven).
- 2022-... Pieter Gijbbers, AI Engineer, Machine learning (TU Eindhoven).

Leadership Experience (continued)

- 2022-2025  Taniya Das, AI Engineer, Deep Learning (TU Eindhoven).
- 2022-2024  Jos van der Velde, AI Engineer, Machine learning (TU Eindhoven).
- 2019-2024  Prabhant Singh, AI Engineer, OpenML core development (TU Eindhoven).
- 2018-2021  Sahitya Ravi, AI Engineer, OpenML core development (TU Eindhoven).

PDEng Student Advisor

- 2018-2020  Yandre Lozano, PDEng, Predictive Maintenance for Smart Buildings (TU Eindhoven).
-  Karthik Srinivasan, PDEng, Preventing Burglaries and Other Incidents (TU Eindhoven).

Other Mentorship

- 2024-...  Aaqib Saeed, Assistant Professor (TU Eindhoven).

Invited Talks

-  SURF Research Day, Hilversum, The Netherlands, May 2026
-  Keynote, International Conference on Discovery Science, Ljubljana, Slovenia, September 2025
-  Santa Fe Institute Workshop on Measuring AI in the World, Santa Fe, March 2025
-  Keynote, Leuven.AI Scientific Meeting, Leuven, May 2024
-  NIST AI Metrology Colloquium, Virtual, Dec 2023
-  VUB AI Lab Seminar. Brussels, October 2023
-  Deep Learning Workshop, Trento, Jun 2023
-  Mathematical Research Data Initiative (MaRDI) Symposium, Berlin, Sep 2022
-  OECD Workshop on AI and the productivity of science, Virtual, Nov 2021
-  Keynote, International Conference on Intelligent Data Engineering and Automated Learning (IDEAL), Virtual, Nov 2021
-  Scalable Data Science Keynote, International Conference on Very Large Data Bases (VLDB), Aug 2021
-  Data-Centric AI event with Andrew Ng, Virtual, Aug 2021
-  Florence Nightingale Symposium, Virtual, Jan 2021
-  Freiburg Machine Learning Lab, Virtual, Dec 2020
-  International FAIR Convergence Symposium, Virtual, Nov 2020
-  ELLIS AutoML Seminar, Virtual, Sep 2020
-  UCI Symposium on Reproducibility in Machine Learning, Virtual, Sep 2020
-  Booking.com Research, Amsterdam, The Netherlands, Jan 2020
-  ECML Workshop on Automated Machine Learning, Wurzburg, Germany, Sep 2019
-  UN Global Summit on AI for Good, Geneva, Switzerland, May 2019
-  Spring Symposium (AI for collaborative data science), AAAI, Stanford, USA, Mar 2019
-  MLOSS Workshop, NeurIPS, Montreal, Canada, Dec 2018
-  AutoML Workshop, PRICAI, Nanjing, China, Aug 2018
-  DEEM Workshop, SIGMOD, Houston, USA, Jun 2018
-  National eScience Symposium, Amsterdam, The Netherlands, Oct 2017
-  Reproducible Machine Learning workshop, ICML, Sydney, Australia, Aug 2017
-  Big data tools for physics and astronomy workshop, Amsterdam, The Netherlands, Jun 2017

Invited Talks (continued)

- Amazon Research, Berlin, Germany, Apr 2017 and Cambridge, UK, Feb 2017
- Challenges in Machine Learning Workshop, NIPS, Barcelona, Spain, Dec 2016
- Dutch Society for Pattern Recognition, Eindhoven, The Netherlands, Nov 2016
- IBM Watson Research Center, New York, USA, Jun 2016
- Machine Learning for High Energy Physics, Lund, Sweden, Jun 2016
- Open Data Science @ Sheffield workshop, Sheffield, UK, Dec 2015
- Horizon Talk, IDA, St Etienne, France, Oct 2015
- Keynote, Statistical Computing (StatComp), Ulm, Germany, Jul 2015
- AutoML Workshop, ICML, Lille, France, Jul 2015
- Keynote, European Conference on Data Analysis (ECDA), Bremen, Germany, Jul 2014

Grants (amounts are funds specifically for my group)

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|-----------|---|
| 2026-... | <ul style="list-style-type: none"> ■ Dutch Science Foundation, Open Science Infrastructure, <i>Hippocampus - Open collaborative infrastructure for AI-Driven Science</i> (PI) €450,000 ■ Dutch Science Foundation, Open Science Infrastructure, <i>OpenML-Next</i> (PI) €170,000 ■ Digital Holland, Dynamic Agentic AI for Plant Technology (co-PI) €150,000 |
| 2025-... | <ul style="list-style-type: none"> ■ EU Horizon Europe, <i>ELLIOT - European Large Open Multi-Modal Foundation Models For Robust Generalization On Arbitrary Data Streams</i> (WPL) €1,560,000 ■ EU Digital Europe, <i>OpenEuroLLM - Open European Family of Large Language Models</i> (WPL) €1,620,000 ■ EU Digital Europe, <i>LLMs4EU - Large Language Models for the European Union</i> (P) €885,000 ■ EU Horizon Europe, <i>MOSAIC - Essential Electronic Components and Systems for our Automated Digital Future in Industry and Mobility</i> (P) €780,000 |
| 2024-... | <ul style="list-style-type: none"> ■ EU Horizon Europe, <i>ALFIE: Assessment of Learning technologies and Frameworks for Intelligent and Ethical AI</i> (P) €330,000 ■ EU Horizon Europe, <i>SYNERGIES: Cooperative, Connected and Automated Mobility</i> (P) €734,000 |
| 2022-2026 | <ul style="list-style-type: none"> ■ Dutch Government, <i>Machine Learning for building renovations</i> (P) €240,000 ■ Dutch Science Foundation, Merian Fund, <i>Digital Twin of a Vertical Farm</i> (Co-PI) €278,000 |
| 2024-2025 | <ul style="list-style-type: none"> ■ EU Digital Europe, <i>European Digital Innovation Hub South Netherlands</i> (P) €120,000 ■ Dutch Science Foundation, Open Science Fund, <i>Automated Machine Learning for all</i> (PI) €50,000 |
| 2022-2025 | <ul style="list-style-type: none"> ■ EU Horizon Europe, <i>AI4Europe</i> (WPL) €506,000 |
| 2020-2024 | <ul style="list-style-type: none"> ■ EU Horizon 2020, <i>Stairway to AI</i> (P) €218,000 ■ ITEA Inno4Health, <i>Continuous monitoring in personal and physical health</i> (P) €517,000 ■ Dutch Science Foundation, TTW, <i>Multi Modal Photochemistry</i> (WPL) €122,000 ■ EU Horizon 2020, <i>TAILOR Network of AI Excellence</i> (WPL) €350,000 (+ managing a €1.5M networking fund) ■ Dutch Science Foundation, <i>SkyHigh: Leveraging AI in Vertical Farming</i> (PL) €300,000 |
| 2019-2021 | <ul style="list-style-type: none"> ■ BOOST, <i>Educational platform for machine learning and medical image analysis</i> (P) €60,000 |
| 2019-2020 | <ul style="list-style-type: none"> ■ Amazon Research Award, <i>The AutoML Gym</i> (PI) \$100,000 |

Grants (amounts are funds specifically for my group) (continued)

2017-2021	■ Dutch Science Foundation, Commit2Data, <i>Dynamic Data Analytics through Automatically Constructed Machine Learning Pipelines</i> (P) €240,000
	■ DARPA, <i>Data Driven Discovery of Models</i> (P) €500,000
2016-2016	■ Microsoft Azure Research Award, <i>A Cloud-Based Platform for AutoML</i> (PI) €40,000
2012-2016	■ Dutch Science Foundation, Free Competition, <i>Massively Collaborative ML</i> (PI) €240,000
2012-2013	■ EU PASCAL Harvest, <i>MLOpen Machine Learning Platform</i> (PI) €30,000

PI: Principal Investigator, PL: Project leader, WPL: Work package leader, P: Participant

Professional Activities

Editorial Boards

2024-...	■ Journal of Data-centric Machine Learning Research (DMLR), Editor-In-Chief
2020-...	■ Journal of Machine Learning Research (JMLR), Action Editor
2022-2024	■ ArXiv.org, Moderator for Machine Learning (CS.LG)
2018-2022	■ Machine Learning Journal (MLJ), Action Editor

Conference Chair

2026	■ NeurIPS Evaluations & Datasets Chair. Conference on Neural Information Processing Systems.
2024	■ AutoML Program Chair. Automated Machine Learning Conference.
2021-2025	■ NeurIPS Datasets & Benchmarks Chair. Conference on Neural Information Processing Systems
2022	■ AutoML Tutorial Chair. Automated Machine Learning Conference
2018	■ DS Program Chair. International Conference on Discovery Science
2016	■ LION General Chair. Learning and Intelligent Optimization Conference
2013	■ ECMLPKDD Demo Chair. European Conference on Machine Learning
2010-2011	■ BeNeLearn Program Chair. Belgian-Dutch Machine Learning Conference

Workshop Chair

2025	■ EurIPS Workshop on the Science of Benchmarking and Evaluation
	■ ICLR Workshop on Future Machine Learning Data Practices and Repositories
2018-2022	■ NeurIPS Workshop on Meta-Learning
2021	■ NeurIPS Workshop on Data-Centric AI
	■ AAAI Workshop on Meta-Learning
2016-2021	■ ICML Workshop on Automatic Machine Learning
2017	■ DALI Workshop on The Data Science Process
	■ ECMLPKDD Workshop on Automatic Machine Learning
2015	■ ECMLPKDD Workshop on Meta-Learning and Algorithm Selection
2014	■ ECAI Workshop on Meta-Learning and Algorithm Selection
2012	■ ECMLPKDD Workshop on Learning from Unexpected Results
	■ ECAI Workshop on Planning to Learn

Professional Activities (continued)

PhD Thesis Examiner

2026	■	Martin Binder (LMU Munich)
	■	Tristan Tomilin (TU Eindhoven)
	■	Bram Grooten (TU Eindhoven)
2025	■	Nabeel Seedat (Univ. of Cambridge)
	■	Zehao Xiao (Univ. of Amsterdam)
	■	Linhao Meng (TU Eindhoven)
	■	Rudy Semola (Univ. Pisa)
2024	■	Sigrid Hellan (Univ. Edinburgh)
	■	Ziqi Wang (TU Delft)
2022	■	Xingchen Ma (KU Leuven)
	■	Herilalaina Rakotoarison (Univ. Paris-Saclay)
	■	Matthias Feurer (Univ. Freiburg)
2021	■	Zhengying Liu (Univ. Paris-Saclay)
	■	Taha Ceritli (Univ. Edinburgh)
	■	Sebastian Flennerhag (Univ. Manchester)
2019	■	Lisheng Sun (Univ. Paris-Saclay)
2017	■	Michel Camilleri (Univ. Malta)
	■	Gitte Vanwickelen (KU Leuven)

PC Committee

2024 - 2025	■	Neural Information Processing Systems (NeurIPS) Senior Area Chair
2022 - 2025	■	Conference on Lifelong Learning Agents (CoLLAs) Senior Reviewer
2022	■	Automated Machine Learning Conference (AutoML)
2012 - 2021	■	International Conference on Machine Learning (ICML) Area chair
2016 - 2020	■	Neural Information Processing Systems (NeurIPS) (Top 10% reviewer)
2019 - 2020	■	Machine Learning and Systems (MLSys)
2012 - 2017	■	European Conference on Machine Learning (ECML-PKDD)
2014 - 2016	■	European Conference on Artificial Intelligence (ECAI)
2016	■	ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)

Education Management

2023-present	■	Education Program Director, Data Science, TU Eindhoven
2022-2023	■	Examination Committee, Data Science & AI Master, TU Eindhoven
	■	Admissions Board, Data Science & AI Master, TU Eindhoven
2015-2023	■	Mentor for TU Eindhoven Master programme
2015-2017	■	Internationalization Coordinator, TU Eindhoven
2014-2018	■	Education Committee, Business Information Systems Master, TU Eindhoven
2014-2017	■	Coach for TU Eindhoven Bachelor School

Journal Articles

- 1 Yildirim, E. C. G., Yildirim, M. O., & **Vanschoren, J.** (2026a). Coresets are more than replay: A data-centric view of continual learning. *Neural Computing and Applications*, *In press*.
- 2 Yildirim, M. O., Yildirim, E. C. G., & **Vanschoren, J.** (2026). Unlocking [CLS] features for continual post-training. *Transactions on Machine Learning Research (TMLR)*, 2026/02.
- 3 Bischl, B., Casalicchio, G., Das, T., Feurer, M., Fischer, S., Gijsbers, P., Mukherjee, S., Müller, A. C., Németh, L., Oala, L., Purucker, L., Ravi, S., van Rijn, J. N., Singh, P., **Vanschoren, J.**, van der Velde, J., & Wever, M. (2025). OpenML: Insights from ten years and more than a thousand papers. *Patterns*, 6(7).
- 4 Cvetkov-Iliev, A., Soulios, V., Xu, L., Abbas, G. M., Kyrou, E., Havinga, L., Hoes, P. J., Loonen, R., & **Vanschoren, J.** (2025). In-depth sensitivity analysis of heating demand and overheating in dutch terraced houses using interpretable machine learning. *Energy and Buildings*, 337, 115611.
- 5 Singh, P., Gijsbers, P., Yildirim, M. O., Gok Yildirim, E. C., & **Vanschoren, J.** (2025). Automated machine learning for unsupervised tabular tasks. *Machine Learning*, 115, Art. 39.
- 6 Gijsbers, P., Bueno, M. L., Coors, S., LeDell, E., Poirier, S., Thomas, J., Bischl, B., & **Vanschoren, J.** (2024). AMLB: an AutoML Benchmark. *Journal of Machine Learning Research*, 25(101), 1–65.
- 7 Moharil, A., **Vanschoren, J.**, Singh, P., & Tamburri, D. (2024). Towards efficient automl: A pipeline synthesis approach leveraging pre-trained transformers for multimodal data. *Machine Learning*, 113, 7011–7053.
- 8 Vettoruzzo, A., Bouguelia, M.-R., **Vanschoren, J.**, Rognvaldsson, T., & Santosh, K. (2024). Advances and challenges in meta-learning: A technical review. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*.
- 9 Weerts, H., Pfisterer, F., Feurer, M., Eggensperger, K., Bergman, E., Awad, N., **Vanschoren, J.**, Pechenizkiy, M., Bischl, B., & Hutter, F. (2024). Can fairness be automated? guidelines and opportunities for fairness-aware AutoML. *Journal of Artificial Intelligence Research*, 79, 639–677.
- 10 Campero Jurado, I., Lorato, I., Morales, J., Fruytier, L., Stuart, S., Panditha, P., Janssen, D. M., Rossetti, N., Uzunbajakava, N., Serban, I. B., Rikken, L., de Kok, M., **Vanschoren, J.**, & Brombacher, A. (2023). Signal quality analysis for long-term ECG monitoring using a health patch in cardiac patients. *Sensors*, 23(4), Art. 2130.
- 11 Celik, B., Singh, P., & **Vanschoren, J.** (2023). Online AutoML: An adaptive AutoML framework for online learning. *Machine Learning*, 112(6), 1897–1921.
- 12 Yildirim, M. O., Gok Yildirim, E. C., Eren, E., Huang, P., Haris, M. P., Kazim, S., **Vanschoren, J.**, Uygun Oksuz, A., & Ahmad, S. (2023). Automated machine learning approach in material discovery of hole selective layers for perovskite solar cells. *Energy Technology*, 11(1).
- 13 Bellido-Jiménez, J. A., Estévez, J., **Vanschoren, J.**, & García-Marín, A. P. (2022). AgroML: An open-source repository to forecast reference evapotranspiration in different geo-climatic conditions using machine learning and transformer-based models. *Agronomy*, 12(3), 656.
- 14 Campero Jurado, I., Fedjajevs, A., **Vanschoren, J.**, & Brombacher, A. (2022). Interpretable assessment of ST-segment deviation in ECG time series. *Sensors*, 22(13), Art. 4919.
- 15 Lijffijt, J., Gkorou, D., Van Hertum, P., Ypma, A., Pechenizkiy, M., & **Vanschoren, J.** (2022). Introduction to the special section on AI in manufacturing: Current trends and challenges. *ACM SIGKDD Explorations*, 24(2), 81–85.
- 16 Rivolli, A., Garcia, L. P., Soares, C., **Vanschoren, J.**, & de Carvalho, A. C. (2022). Meta-features for meta-learning. *Knowledge-Based Systems*, 240, 108101.

- 17 Zhang, C., **Vanschoren, J.**, van Wissen, A., Lakens, D., de Ruyter, B., & IJsselsteijn, W. A. (2022). Theory-based habit modeling for enhancing behavior prediction in behavior change support systems. *User Modeling and User-Adapted Interaction*, 23.
- 18 Balázs, C., van Beekveld, M., Caron, S., Dillon, B. M., Farmer, B., Fowlie, A., Garrido-Merchán, E. C., Handley, W., Hendriks, L., Jóhannesson, G., Mamužić, J., Martinez, G., Scott, P., Ruiz de Austri, R., Searle, Z., Stienen, B., **Vanschoren, J.**, & White, M. (2021). A comparison of optimisation algorithms for high-dimensional particle and astrophysics applications. *Journal of High Energy Physics*, 2021(5), 1–46.
- 19 Celik, B., & **Vanschoren, J.** (2021). Adaptation strategies for automated machine learning on evolving data. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 43(9), 3067–3078.
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